
INTRODUCTION TO EMBEDDED SYSTEMS AND IOTs

Lab Name: Embedded Light Flow System

Lab Number: 1

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Student Name: Nicole Adelson

Student ID: 703226743

INTRODUCTION OF THE LAB AND REPORT

THE PURPOSE OF THIS LAB WAS TO DEVELOP AN EMBEDDED SYSTEM THAT DETECTS AND ROTATES TOWARDS THE DIRECTION OF MAXIMUM LIGHT INTENSITY USING THE ARDUINO UNO BOARD. A SERVO MOTOR, WITH A PHOTORESISTOR CONNECTED TO THE ROTATING ARM, INITIALLY SWEEPS IN A 180-DEGREE MOTION TO COLLECT LIGHT INTENSITY READINGS. BASED ON THESE REAL-TIME DATA, THE SYSTEM CALCULATES THE DIRECTION OF THE MOST INTENSE LIGHT AND ADJUSTS THE ARM ATTACHMENT ACCORDINGLY. THE SYSTEM ALSO INCORPORATES USER-DEFINED PARAMETERS, INCLUDING SCAN DELAY INTERVALS AND SENSITIVITY THRESHOLDS, WHICH CAN BE MODIFIED THROUGH SERIAL INPUT AT RUNTIME. FURTHERMORE, THE SYSTEM RESPONDS TO RESET INPUT THROUGH A PUSH BUTTON OR CHANGES IN AMBIENT LIGHT THROUGH SENSITIVITY SETTINGS DEFINED BY THE USER. USING THE ARDUINO IDE, THE MICROCONTROLLER WAS PROGRAMMED AND IMPLEMENTED SERIAL COMMUNICATION TO LOG REAL-TIME PHOTORESISTOR READINGS AND DEBUGGING STATEMENTS. THIS SYSTEM IS AN EFFECTIVE PROTOTYPE FOR APPLICATIONS SUCH AS SOLAR ENERGY INSTALLATIONS, WHICH RELY ON THE DETECTION OF MAXIMUM DIRECT SUNLIGHT IN ORDER TO HELP THE SOLAR TRACKERS POSITION THE PANELS FOR OPTIMAL EFFICIENCY.

MATERIALS, DEVICES AND INSTRUMENTS

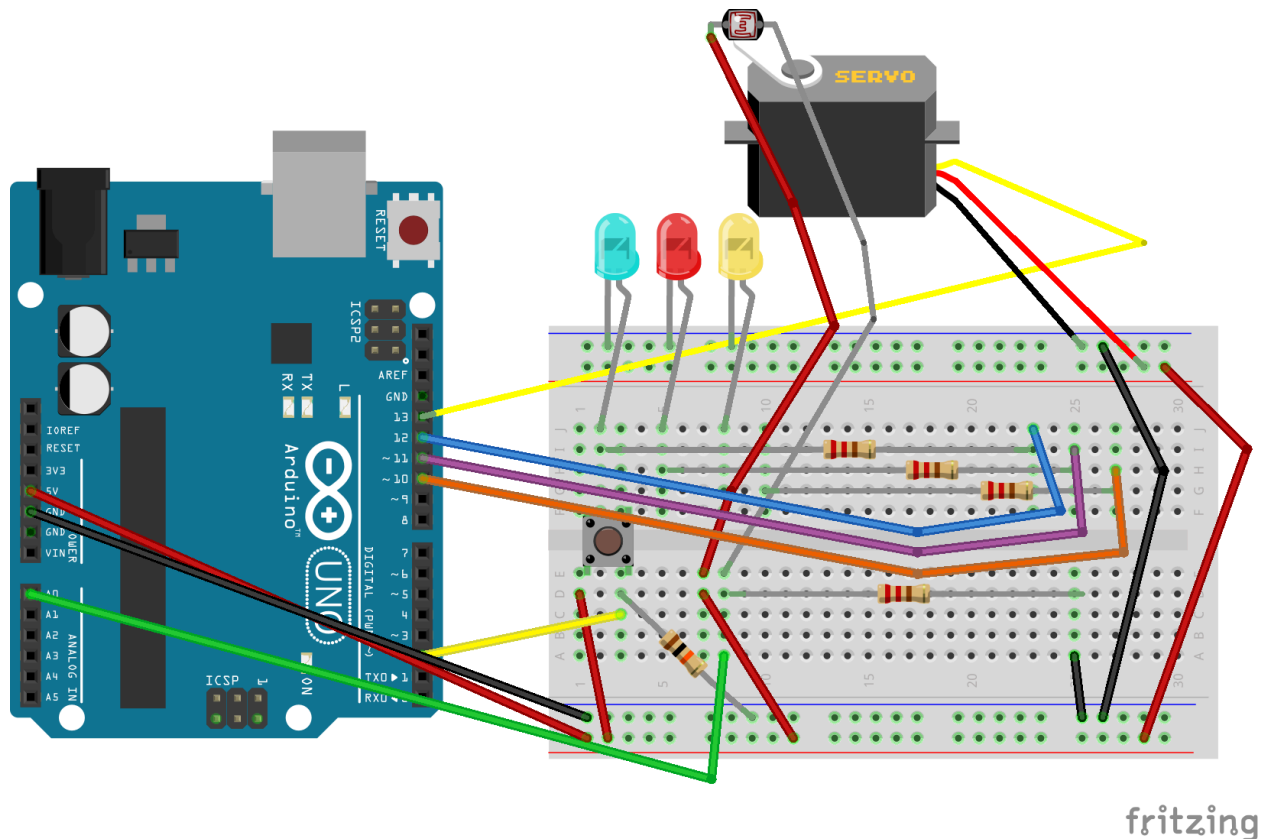
Materials:

- 1 Breadboard
- 1 Servo Motor (with arm attachment)
- 3 LEDs (blue, red, yellow)
- 4 x 220 Ω resistors
- 1 x 10k Ω resistor
- 13 male-to-male jumper wires
- 2 female-to-male jumper wires

Devices:

- Arduino UNO
- HP Laptop with Arduino IDE

SCHEMATICS



LAB METHODS AND PROCEDURE

This light-tracking system was implemented on a breadboard using an Arduino Uno microcontroller. The servo motor was connected to PWM pin 9, the push-button for rescans connected to PWM pin 2, and the three LEDs connected to pins 4 (blue), 5 (yellow), and 6 (red) with 220 Ω resistors. The photoresistor was mounted on the servo arm using electrical tape to face outward toward the light source. A voltage divider circuit was developed by connecting one leg to the 5V pin the other leg to ground through a 10k Ω resistor, and connecting the junction to analog pin A0. The push-button was connected to the 10k Ω pull-up resistor, and events were tracked through an external interrupt on pin 2. Automatic scanning was maintained in the main loop using the `millis()` function for timing. Additionally, a boolean flag was set inside the ISR to signal a rescan, which allowed the main loop to control the servo movements, LED updates, and scanning periods

without conflicts. Serial communication was also initialized to display debugging information and log photoresistor readings.

During each scan, the servo sweeps from 0 degrees to 180 degrees in 5-degree increments, pausing briefly at each step to allow the photoresistor readings to stabilize. At each step, multiple photoresistor readings are averaged to minimize noise. The system continuously records the values from the photoresistor to determine the angle with the maximum light intensity. Once identified, the servo rotates to this position, aligning the arm with the brightest light source. During this process, the yellow LED turns on to indicate active scanning, the blue LED turns on when the system is ready and after a full scan, and the red LED illuminates if there is an interrupt either due to the time elapsed reaching the user-defined interval or the change in light intensity exceeding the user-defined sensitivity threshold.

TROUBLE IN THE LAB AND HOW DID YOU SOLVE IT

There were various challenges in hardware and software integration for the light-tracking system. One key hardware issue was guaranteeing that the photoresistor remained stable during operation, as movement would lead to inconsistent readings. This was resolved by securing it to the servo arm with electrical tape. The servo arm itself also presented challenges, initially failing to rotate consistently due to conflicting TimerOne interrupts. This functionality was temporarily removed to restore proper motion and address any coinciding issues. Additionally, the connection of the female-to-male jumper wires to the photoresistor were prone to disconnection, and the red LED failed to illuminate, both of which were addressed by carefully securing the components and physical adjustments, such as electrical tape around the jumper wire connection.

From the software perspective, accurately tracking the light source required more than one reading per servo angle and averaging the values to reduce noise, while also ignoring substantial jumps in intensity to ensure stability. The manual rescans by pressing the pushbutton initially did not register, which required proper debouncing of 200ms alongside the rescan flag in the ISR. Each scan was set to start at 0 degrees upon setting the rescan flag and all of the readings from the previous sweep were wiped in order to demonstrate a sufficient reset of the system. These hardware and software adjustments allowed the system to achieve consistent, responsive operation.

RESULTS AND CONCLUSION

The system was able to successfully track the brightest light source after sweep sessions. The servo accurately rotated towards the maximum photoresistor reading, and rescans were effectively instantiated by the timer every 5 seconds and manually by pressing the push-button. The LED feedback worked as intended, with the yellow LED illuminated during the scanning period, the blue LED turned on when the system was ready and when there was a successful full scan, and the red LED displaying that the system had exceeded its set runtime or detected a light interval change that exceeded the defined sensitivity threshold. The serial outputs were able to confirm the maximum intensity angle and the corresponding photoresistor values, demonstrating reliable

analysis among several tests. The system was able to respond consistently, with varying ambient light levels.

The successful design and implementation of this light-tracking system shows the effective application of sensor input, servo control, user-defined parameters, and interrupts in an applicable Arduino-based project. The use of event-driven interrupts for rescans based on light change and push-button input enabled flexibility between automatic and manual scans, while the dynamic LED feedback displayed the system states easily for users. Despite its simplicity, the system adapts crucial principles in embedded design, including minimal interrupts, clear feedback, and reliable infrastructure. This project demonstrates the use of microcontrollers for real-time environmental tracking, and provides a sturdy foundation for further applications.